

MedMind AI at #SMM4H-HeaRD 2026: Data Extraction and Generation Using Prompt Engineering and Structured Outputs (Tasks 1–6)

Aatish Pradhan

Independent Researcher

aatishpradhan11@gmail.com

Brian Habersberger

Independent Researcher

habersberger@gmail.com

Abstract

Six tasks from the SMM4H–HeaRD 2026 workshop were addressed with task-specific large-language-model (LLM) pipelines relying on prompt engineering, strict structured (JSON) responses, and deterministic rule sets. The pipelines utilize no task-specific fine-tuning and can be adapted across diverse clinical and social media data. This study demonstrates that general-purpose LLMs (**gpt-5.4-mini** and **gpt-5.4**) can accurately extract and classify crucial health information when constrained by strict output schemas. Notably, our hybrid approach achieved the best overall performance among all participating systems for Task 2 (Insomnia Detection).

1 Introduction

Social-media mining for health (SMM4H) tasks conventionally employ domain-specific BERT variants or sequence-to-sequence fine-tuning. These methods incur substantial computational overhead and require extensive supervised datasets.

Recent large language models (LLMs) mitigate this via constrained inference, compelling the model to emit syntactically valid JSON that follows a user-supplied schema. Embedding key decision logic within this schema often circumvents the need for explicit task-specific fine-tuning (Pradhan et al., 2025).

We evaluate this approach across six heterogeneous tasks for SMM4H–HeaRD 2026 (Lopez-Garcia et al., 2026): multilingual adverse drug events, insomnia extraction in MIMIC-III (Johnson et al., 2016), flu vaccination status (Xu et al., 2024), Dial2Note Subjective, Objective, Assessment, and Plan (SOAP) generation (Mianroodi et al., 2025), patient metadata detection (Klein et al., 2025), and Tumor, Node, and Metastasis (TNM) pathology staging (Kefeli et al., 2024). The main contributions are: 1. A schema-constrained framework (**gpt-5.4-mini**, **gpt-5.4**) leveraging targeted prompt

engineering to address six distinct tasks without fine-tuning. 2. Demonstrating that hybrid LLM-regex pipelines improve clinical extraction fidelity (Task 2). 3. Analyzing discrepancies between ground-truth labels and LLM predictions through the lens of strict entity-relationship schemas.

2 Methodology

Structured output schemas guide the model to generate responses that are subsequently parsed or processed using rule-based label assignment. By formalizing both the entities and their relationships to source evidence through this schema, the LLM can reliably disentangle complex narratives and deliver machine-readable extractions aligned with task guidelines.

2.1 Architecture

- **Input:** Raw text from the provided social media or clinical datasets.
- **Prompt Assembly:**
 - (SYSTEM) high-level domain instruction;
 - (USER) the cleaned document;
 - (SCHEMA) strict JSON output definition.
- **Model Call:** The majority of tasks utilize **gpt-5.4-mini** at temperature 0.0 with strict JSON structured outputs enforced. For the multilingual dataset (Task 1), the larger **gpt-5.4** model was deployed to handle the increased complexity of cross-lingual reasoning. Average inference time per document was approximately 1.2 seconds. Across the entire project lifecycle (totaling 132,177 requests across training, validation, and evaluation), the empirical API cost averaged \$0.0024 per sample.

- **Validation:** Rule-based heuristics were applied to the structured LLM responses to map findings to final task labels.

Prompt architectures and schema constraints were iteratively refined during the development phase, with final configurations selected based on their empirical performance on the validation sets.

2.2 Task 1: Multilingual ADE Detection

Task 1 centers on extracting adverse drug effects across multiple languages. Review of the training data revealed language- and source-specific annotation guidelines. For example, in the German dataset, dietary supplements and herbal/phytomedicines were annotated as pharmaceutical drugs, requiring an expanded definition within the prompt. Furthermore, brand name disambiguation was critical; in the Japanese dataset, "ル ル" (Lulu) could refer to either a common cold medicine or a pet name. Language-specific system prompts were engineered to address these observed exceptions, improving the resolution of adverse drug events without post-hoc rule processing.

2.3 Task 2: Insomnia in Clinical Notes

Task 2 required identifying diagnostic criteria and medication spans in clinical notes. The architecture incorporated a hybrid approach: a deterministic regex-based annotator first flagged criteria and canonical drug names. This draft was passed to the LLM for contextual review to make potential corrections.

2.4 Task 3: Flu Vaccine Status on Twitter

This classification task differentiates flu vaccination intentions, actions, and test results. The prompt included the tweet's `created_at` timestamp to provide temporal grounding, ensuring the LLM could distinguish between current and previous flu seasons.

2.5 Task 4: Dial2Note Generation

Task 4 focuses on generating structured SOAP notes from clinical dialogue. Strict grounding rules were provided in the prompt, steering the model from fabricating demographics or provider names not explicitly present in the source text.

2.6 Task 5: Patient Metadata Detection

The objective is to perform a binary classification distinguishing online article sentences that report

patient-related metadata from those that do not. The schema was simplified to a binary metadata flag, successfully preventing the model from falling into negative classification traps.

2.7 Task 6: TNM Extraction

This task extracts T, N, and M cancer stages from pathology reports. Engineered search directives within the prompt forced the model to actively scan for distant organ involvement, preventing it from reflexively defaulting to an M0 stage.

3 Structured LLM Outputs as a Lens for Task Definition

High-performance manual prompt engineering, coupled with a strictly enforced structured response format, improves accuracy while revealing subtle variations in the interpretation of clinical texts.

Task 1 Validation Insights: Handling multilingual extraction highlighted how schemas expose cultural task ambiguities. Applying a uniform set of English-centric extraction rules misclassified phytomedicines in the German dataset (where they are often annotated as pharmaceutical drugs). Moving these constraints into language-specific system prompts allowed the LLM to apply rules with contextual nuance.

Task 2 Validation Insights: Extracting evidence spans for insomnia highlighted the challenges of rigid token matching. The LLM naturally extracts broader semantic contexts (e.g., a full explanatory sentence) rather than the minimal specific phrases preferred by annotators. Utilizing a hybrid deterministic-LLM approach proved essential for balancing semantic understanding with strict extraction boundaries.

Task 3 Validation Insights: Initial LLM outputs frequently struggled to distinguish between past actions, current actions, and future intentions (e.g., misclassifying future intentions as "Currently-Vaccinated"). Dynamically injecting the `created_at` timestamp anchored the LLM's reasoning in a concrete temporal context.

Task 4 Validation Insights: When generating SOAP notes, the LLM initially hallucinated clinical content. This discrepancy was traced back to an over-reliance on few-shot examples in the prompt, which inadvertently anchored the model to specific formatting patterns from the training data (which contained baseline dataset hallucinations). Removing few-shot examples and enforcing

explicit instruction-based grounding yielded vastly more faithful outputs.

Task 5 Validation Insights: The interaction between the JSON schema and reasoning revealed a phenomenon we term "trap commitment bias." We formally define this as the phenomenon where an LLM, once forced by a schema to explicitly categorize a challenging negative example (the "trap," e.g., virology methodology), anchors its subsequent token generation to that negative classification, causing it to ignore secondary positive signals within the same text. This differs from standard anchoring bias and contextualizes the model's resistance to correction noted in the baseline system (Klein et al., 2025). Simplifying to a boolean output resolved this discrepancy.

Task 6 Validation Insights: Extracting TNM cancer stages revealed a strong LLM bias toward majority classes, particularly assuming no metastasis (*M0*). Engineering strict search directives within the prompt forced the model to explicitly document its search for distant metastases in a chain-of-thought field prior to emitting the final stage.

4 Results

Table 1 summarizes the performance of our system across all six tasks on the official SMM4H-HeaRD 2026 test sets.

Our strictly schema-constrained approach achieved highly competitive results, most notably yielding the best overall performance among all participating systems for Task 2 (Insomnia Detection). In Subtask 1, our hybrid LLM-regex pipeline achieved an F1-score of 0.8649. In Subtask 2, our system secured an Exact Match span extraction score of 0.7170 and a Label Classification F1 of 0.8764, substantially outperforming the respective task means. In Task 1, our unweighted macro F1-score of 0.6518 exceeded the participant mean. Temporal grounding in Task 3 enabled us to surpass participant means across both subtasks. For Task 6, our system nearly achieved perfect extraction on the primary Test Set 1. The system performed slightly above average for Task 4, and underperformed on Task 5.

5 Discussion

The performance metrics across the test sets validate the core hypothesis that schema-constrained, general-purpose LLMs can effectively compete

with, and occasionally outperform, task-specific fine-tuned models in clinical NLP.

The Superiority of Hybrid Systems for Span Extraction: The system's best-in-class performance on Task 2 underscores a known limitation of pure LLM approaches: while they excel at semantic reasoning, they frequently fail at precise, token-level boundary extraction. The success of our pipeline resulted from delegating rigid span boundary identification to a deterministic regex annotator and utilizing the LLM solely as a contextual filter.

Limitations of Zero-Shot Schemas in Specialized Domains: Conversely, the underperformance on Task 5 (Metadata Detection) highlights a challenge of this approach. Although simplifying the schema to mitigate "trap commitment bias" improved our validation metrics, reasoning against our prompt was insufficient to represent the complex relationship between the epidemiological data and annotations.

The Necessity of Grounded Reasoning: Tasks 3 and 6 demonstrated that LLMs are highly susceptible to their pre-training priors without explicit grounding. Injecting temporal anchors (Task 3) and forcing the model to explicitly document its search for metastasis in a chain-of-thought field (Task 6) successfully prevented the model from defaulting to majority classes.

Evaluation Metrics vs. Clinical Factuality: In Task 4, our prompt enforced strict clinical grounding, explicitly prohibiting the model from fabricating demographic details not present in the source dialogues. However, the reference "ground truth" notes frequently contain these hallucinated demographic details resulting from the dataset's dialogue-to-notes synthesis process. Standard automated metrics like BLEU penalize strict clinical factuality when rewarding baseline hallucinations.

Limitations

A primary limitation of this work is the highly task-specific nature of the prompts and schemas. While the underlying strategy of schema-constrained inference is highly adaptable and requires very low computational overhead to initiate, the specific pipelines and logical rule sets described here do not generalize zero-shot to new tasks without manual dialogue with the model, adjustment of prompts, and schema redesign. Additionally, the approach relies on proprietary, closed-weight models (GPT-

Task	Evaluation Target	Primary Metric	Val (Ours)	Test (Ours)	Test Mean	Organizer Baseline
1	Multilingual ADE	Macro F1	0.5923	0.6518	0.5465	-
2, ST1	Binary Classification	F1-score	0.9091	0.8649	0.6805	0.9300
2, ST2	Label Classification	Micro F1	0.8000	0.8764	0.5888	0.9300
2, ST2	Span Extraction	Exact Match F1	0.5974	0.7170	0.3129	-
3, ST1	Flu Vaccination	Micro F1	0.8148	0.8932	0.8499	0.8700
3, ST2	Flu Test Result	Micro F1	0.9704	0.9326	0.9071	0.8700
4	Text Generation	Average Score	0.5203	0.4800	0.4700	0.6082
5	Metadata Detection	F1-score	0.5083	0.5860	0.7290	0.7760
6 (TS1)	TNM Staging (T)	Macro F1	0.8438	0.9960	-	0.9920
6 (TS1)	TNM Staging (N)	Macro F1	0.8483	0.9980	-	0.7830
6 (TS1)	TNM Staging (M)	Macro F1	0.7543	0.9970	-	0.7960
6 (TS2)	TNM Staging (T)	Macro F1	-	0.8490	-	0.4540
6 (TS2)	TNM Staging (N)	Macro F1	-	0.8650	-	0.5910
6 (TS2)	TNM Staging (M)	Macro F1	-	1.0000	-	0.5540

Table 1: Consolidated Performance Matrix. Validation scores reflect internal evaluations prior to submission. "Test Mean" represents the average performance of participating systems, while "Organizer Baseline" represents the organizer-provided reference models. Bold values indicate the highest score across our system, the participant mean, and the baseline.

5.4 series), limiting full transparency into the model’s underlying pre-training priors.

6 Conclusion

This work introduced a framework using structured output schemas combined with optimized manual prompt strategies to extract vital health information from unstructured texts. By embedding entity-relationship frameworks into schema-constrained prompts, researchers can guide LLMs to maintain a stable chain of reasoning and achieve highly competitive task performance without fine-tuning.

References

- Alistair Johnson, Tom Pollard, and Roger Mark. 2016. [MIMIC-III Clinical Database](#). *PhysioNet*. Version 1.4.
- J. Kefeli, J. Berkowitz, J. M. Acitores Cortina, and 1 others. 2024. [Generalizable and automated classification of TNM stage from pathology reports with external validation](#). *Nature Communications*, 15:8916.
- Ari Z. Klein, Davy Weissenbacher, Karen O’Connor, Amir Elyaderani, Ivan Flores Amaro, Takeshi Onishi, Su Golder, Kaelen Spiegel, Matthew Scotch, and Graciela Gonzalez-Hernandez. 2025. [Detection of patient metadata in published articles for genomic epidemiology using machine learning and large language models](#). *medRxiv*.
- Guillermo Lopez-Garcia, Jose Miguel Acitores Cortina, Jacob Berkowitz, Joey Chan, Ganesh Chandrasekar, Sumon Kanti Dey, Ivan Flores Amaro, Fernando Gallego, Lauren Gryboski, Ari Z Klein, Martin Krallinger, Salvador Lima-López, Tomohiro Nishiyama, Lisa Raithel, Ahmad Rezaie Mianroodi, Roland Roller, Judith Rosell, Frank Rudzicz, Abeer Sarker, and 8 others. 2026. Overview of the 11th Social Media Mining for Health (#SMM4H) and Health Real-World Data (HeaRD) Shared Tasks at ACL 2026. In *Proceedings of the 11th Social Media Mining for Health (#SMM4H) and Health Real-World Data (HeaRD) Workshop and Shared Tasks*. Association for Computational Linguistics.
- Ahmad Rezaie Mianroodi, Amirali Rezaie, Niko Grisel Todorov, Cyril Rakovski, and Frank Rudzicz. 2025. [Medsynth: Realistic, synthetic medical dialogue-note pairs](#). *Preprint*, arXiv:2508.01401.
- Aatish Pradhan, Brian M. Habersberger, James H. Wade, Denver Dsouza, and Nihal Paul. 2025. [LLM pros at SMM4H-HeaRD 2025 data extraction using prompt engineering and structured outputs \(task 1, 2, 3, 4, 5, 6\)](#). In *Workshop Proceedings of the 19th International AAAI Conference on Web and Social Media*. Workshop: #SMM4H-HeaRD 2025: Joint 10th Social Media Mining for Health and Health Real-World Data Workshop and Shared Tasks.
- D. Xu, G. L. García, K. O’Connor, H. Holston, A. Z. Klein, I. F. Amaro, M. Scotch, and G. Gonzalez-Hernandez. 2024. Mining social media data for influenza vaccine effectiveness using a large language model and chain-of-thought prompting. In *AMIA Annual Symposium Proceedings*, pages 1404–1413. American Medical Informatics Association.

A Appendix: Detailed Pipeline Elements

A.1 Task 1: Language-Specific Validation Performance

To contextualize the macro F1-score, the model's performance on the validation split for Task 1 is broken down by language below:

- **English (F1-en):** 0.6577
- **French (F1-fr):** 0.7059
- **German (F1-de):** 0.5577
- **Russian (F1-ru):** 0.5650
- **Overall (Macro F1):** 0.5923

A.2 Task 2: Regex-Based Annotator Details

To overcome the LLM's tendency to extract overly broad semantic spans, a deterministic Regex annotator was deployed as a first pass. The annotator used a predefined dictionary of exact-match string arrays covering all Group A and Group B canonical drug names, as well as highly specific diagnostic criteria phrasing mapped directly from the MIMIC-III training data. The outputs of this deterministic pass were then ingested by the LLM, which acted purely as a pruning filter to eliminate false-positive span extractions based on the surrounding sentence context.

A.3 Task 1: Representative Language-Specific Prompts and Schema

To address source-specific annotation guidelines, the systemic domain rules were tailored per language. Below are excerpts demonstrating how the pharmaceutical definition was adapted between the English and German prompts prior to schema enforcement:

English System Prompt Excerpt:

```
// SYSTEM PROMPT
You are an expert at detecting Adverse Drug Events (
ADEs) in English social media posts from
Twitter/X.
DEFINITION: An ADE is an unwanted, harmful, or
unpleasant health effect that someone ACTUALLY
EXPERIENCED as a result of taking a
pharmaceutical drug or medication.
Rule 1: A pharmaceutical drug or OTC medicine is
mentioned or clearly implied (NOT supplements,
vitamins, coffee, alcohol, energy drinks, food,
or herbal remedies).
```

German System Prompt Excerpt:

```
// SYSTEM PROMPT
You are an expert at detecting Adverse Drug Events (
ADEs) in German patient forum posts.
FOR GERMAN DATA - EXPANDED PHARMACEUTICAL DEFINITION
:
All of the following count as drugs/medications:
```

- Prescription drugs and OTC medicines
- Dietary supplements and vitamins (vitamin D, magnesium, omega-3, etc.)
- Herbal medicines and phytomedicines (black cohosh/ Traubensilberkerze, St. John's Wort/ Johanniskraut, valerian, agnus castus)
- Hormone preparations including HRT (Estragel, Famenita, Zoladex, Livial, hormonal IUD)

JSON Schema:

```
{
  "name": "adverse_effect_detection",
  "strict": true,
  "schema": {
    "type": "object",
    "required": [
      "unique_id",
      "language",
      "english_version",
      "reasoning",
      "adverse_drug_event_mentioned",
      "verbatim_ADE_mention_list"
    ],
    "additionalProperties": false,
    "properties": {
      "unique_id": {
        "type": "string",
        "description": "The unique identifier as provided in the text"
      },
      "language": {
        "type": "string",
        "description": "The language of the text",
        "enum": [
          "English", "German", "French", "Russian", "Mandarin", "Japanese", "Persian", "Other"
        ]
      },
      "english_version": {
        "type": "string",
        "description": "English translation of the text. Repeat the original if already in English."
      },
      "reasoning": {
        "type": "object",
        "description": "Step-by-step reasoning fields to evaluate before making the final classification.",
        "required": [
          "drug_mentioned",
          "drug_name_if_any",
          "is_pharmaceutical",
          "symptom_described",
          "symptom_summary",
          "who_experienced",
          "is_personal_experience",
          "temporal_relationship",
          "is_therapeutic_effect",
          "classification_rationale"
        ]
      },
      "additionalProperties": false,
      "properties": {
        "drug_mentioned": {
          "type": "boolean",
          "description": "Is any drug or medication mentioned or clearly implied?"
        },
        "drug_name_if_any": {
          "type": "string",
          "description": "Name of the drug(s) mentioned, or 'none' if no drug is mentioned."
        },
        "is_pharmaceutical": {
          "type": "boolean",
          "description": "Is the substance a pharmaceutical drug or OTC medicine? False for coffee, alcohol, food, supplements (unless German data where supplements count)."
        },
        "symptom_described": {
          "type": "boolean",
          "description": "Is an unintended health
```


used to treat severe or debilitating insomnia ...

Triazolam: sold under the brand name Halcion among others, is a central nervous system (CNS) depressant tranquilizer of the triazolobenzodiazepine (TBZD) class...

Zaleplon: sold under the brand name Sonata among others, is a sedative and hypnotic which is used to treat insomnia. It is a nonbenzodiazepine or Z-drug of the pyrazolopyrimidine class...

Zolpidem: sold under the brand name Ambien among others, is a medication primarily used for the short-term treatment of sleeping problems...

</Drugs in Drug Group A>

<Drugs in Drug Group B>

Acamprosate: sold under the brand name Campral, is a medication which reduces alcoholism cravings.

Alprazolam: sold under the brand name Xanax among others, is a fast-acting, potent tranquilizer of moderate duration within the triazolobenzodiazepine group...

Clonazepam: sold under the brand name Klonopin, Rivotril, Paxam, & others, is a benzodiazepine medication used to prevent and treat anxiety disorders, seizures, bipolar mania...

Clonidine: sold under the brand name Catapres among others, is an α_2 -adrenergic receptor agonist medication used to treat high blood pressure, attention deficit hyperactivity disorder (ADHD) ...

Diazepam: sold under the brand name Valium among others, is a medicine of the benzodiazepine family that acts as an anxiolytic

Diphenhydramine: sold under the brand name Benadryl, Unisom, Nytol, & others, is an antihistamine and sedative.

Doxepin: sold under the brand name Sinequan, Quitaxon, Aponal, & others, is a medication belonging to the tricyclic antidepressant (TCA) [10] class...

Gabapentin: sold under the brand name Neurontin among others, is an anticonvulsant medication primarily used to treat neuropathic pain...

Hydroxyzine: sold under the brand names Atarax and Vistaril among others, is an antihistamine medication

Lorazepam: sold under the brand name Ativan among others, is a benzodiazepine medication

Melatonin: sold under the brand name Circadin, Slenyto, & others, is a naturally occurring hormone produced in the brain that is also used as a dietary supplement...

Mirtazapine: sold under the brand name Remeron among others, is an atypical tetracyclic antidepressant...

Olanzapine: sold under the brand name Zyprexa among others, is an atypical antipsychotic primarily used to treat schizophrenia and bipolar disorder

Quetiapine: sold under the brand name Seroquel among others, is an atypical antipsychotic medication used in the treatment of schizophrenia, bipolar disorder, bipolar depression, and major depressive disorder

Trazodone: sold under the brand name Desyrel, Trittico, & others, is an antidepressant medication used to treat major depressive disorder, anxiety disorders, and insomnia.

</Drugs in Drug Group B>

// USER PROMPT

Analyze the following anonymized clinical note and identify all evidence relevant to the insomnia diagnostic criteria defined in the system prompt.

Instructions:

1. Read the note carefully.
2. For each criterion under "Difficulty Sleeping" (Definition 1) that is supported by the note, add an entry to 'difficulty_sleeping_identified_criteria' with:
 - the matching 'criterion' (from the allowed enum),

- a minimal 'citation' quoted from the note (use "..." to indicate omitted text within a citation),
 - a 'regular_expression_matching_citation' that uniquely matches that citation in the original note.
3. Do the same for each criterion under "Daytime Impairment" (Definition 2) in 'daytime_impairment_identified_criteria'.
 4. For every Group A drug (primary insomnia medication) mentioned as prescribed/administered to the patient, add an entry to 'group_a_drugs' with the drug name (generic form from the enum), citation, and regex.
 5. For every Group B drug (secondary medication) mentioned as prescribed/administered, add an entry to 'group_b_drugs' likewise. Recognize brand names and map them to the generic enum value.
 6. Only include criteria and drugs that are explicitly supported by the note. Do not infer or speculate. If nothing applies for a category, return an empty array.
 7. Citations must be copied verbatim from the note (aside from "..." for elision) so they can be located by string/regex search.
 8. Return ONLY the JSON object matching the required schema - no prose, no markdown.

JSON Schema:

```
{
  "name": "clinical_notes_criteria",
  "strict": true,
  "schema": {
    "type": "object",
    "properties": {
      "reasoning": {
        "type": "string",
        "description": "Brief step-by-step reasoning grounded in the note before producing structured output."
      },
      "difficulty_sleeping_identified_criteria": {
        "type": "array",
        "items": {
          "type": "object",
          "required": [
            "criterion",
            "citation",
            "regular_expression_matching_citation"
          ],
          "additionalProperties": false,
          "properties": {
            "criterion": {
              "type": "string",
              "enum": [
                "Trouble initiating or maintaining sleep",
                "Waking up earlier than desired",
                "An explicit mention of insomnia"
              ]
            },
            "citation": {
              "type": "string",
              "description": "Cite a concise and minimal amount of text (indicate sub- and partial sentence sampling via '...') supporting the identified criterion. For example, 'Reporting being unable to sleep...' is a concise and minimal citation supporting 'Trouble initiating or maintaining sleep'"
            },
            "regular_expression_matching_citation": {
              "type": "string",
              "description": "A regular expression that can be used to uniquely match the citation"
            }
          }
        }
      },
      "daytime_impairment_identified_criteria": {
```



```

"type": "array",
"items": {
  "type": "object",
  "required": [
    "criterion",
    "citation",
    "regular_expression_matching_citation"
  ],
  "additionalProperties": false,
  "properties": {
    "criterion": {
      "type": "string",
      "enum": [
        "Fatigue or malaise",
        "Impaired attention, concentration, or memory",
        "Impaired social, family, occupational, or academic performance",
        "Mood disturbance or irritability",
        "Daytime sleepiness",
        "Behavioral problems such as hyperactivity, impulsivity, or aggression",
        "Decreased motivation, energy, or initiative",
        "Proneness to errors or accidents",
        "Concerns or dissatisfaction with sleep",
        "An explicit mention of insomnia"
      ]
    },
    "citation": {
      "type": "string",
      "description": "Cite a minimal amount of text (indicate sub- and partial sentence sampling via '...') supporting the identified criterion"
    },
    "regular_expression_matching_citation": {
      "type": "string",
      "description": "A regular expression that can be used to uniquely match the citation"
    }
  }
},
"group_a_drugs": {
  "type": "array",
  "description": "List of drugs classified under group A.",
  "items": {
    "type": "object",
    "properties": {
      "drug_name": {
        "type": "string",
        "enum": [
          "Estazolam",
          "Eszopiclone",
          "Flurazepam",
          "Lemborexant",
          "Quazepam",
          "Ramelteon",
          "Suvorexant",
          "Temazepam",
          "Triazolam",
          "Zaleplon",
          "Zolpidem"
        ]
      }
    },
    "citation": {
      "type": "string",
      "description": "Cite a minimal amount of text (indicate sub- and partial sentence sampling via '...') supporting the identified drug"
    },
    "regular_expression_matching_citation": {
      "type": "string",
      "description": "A regular expression that can be used to uniquely match the citation"
    }
  }
},
"group_b_drugs": {
  "type": "array",
  "description": "List of drugs classified under group B.",
  "items": {
    "type": "object",
    "properties": {
      "drug_name": {
        "type": "string",
        "enum": [
          "Acamprosate",
          "Alprazolam",
          "Clonazepam",
          "Clonidine",
          "Diazepam",
          "Diphenhydramine",
          "Doxepin",
          "Gabapentin",
          "Hydroxyzine",
          "Lorazepam",
          "Melatonin",
          "Mirtazapine",
          "Olanzapine",
          "Quetiapine",
          "Trazodone"
        ]
      },
      "citation": {
        "type": "string",
        "description": "Cite a minimal amount of text (indicate sub- and partial sentence sampling via '...') supporting the identified drug"
      },
      "regular_expression_matching_citation": {
        "type": "string",
        "description": "A regular expression that can be used to uniquely match the citation"
      }
    },
    "required": [
      "drug_name",
      "citation",
      "regular_expression_matching_citation"
    ],
    "additionalProperties": false
  }
},
"required": [
  "reasoning",
  "difficulty_sleeping_identified_criteria",
  "daytime_impairment_identified_criteria",
  "group_a_drugs",
  "group_b_drugs"
],
"additionalProperties": false
}
}

```

A.5 Task 3: Flu Vaccine Prompts and Schema Prompt Structure (Subtask 1: Flu Vaccination Status):

// SYSTEM PROMPT

You are an expert annotator for a health informatics shared task. Your job is to classify social media posts (tweets) according to the author's self-reported flu vaccination status.

Classify each tweet into exactly one of the following five categories:

- **Currently-Vaccinated:** The author describes a **SPECIFIC, COMPLETED** vaccination event in the current or most recent flu season. Look for past-tense or just-completed language: "I got my flu shot", "just got vaccinated", "got my flu shot today", "I received my flu vaccine last week". The key requirement is that the author confirms the shot already happened during this season.
- **Currently-Unvaccinated:** The author explicitly or implicitly indicates they have **NOT** received and do not intend to receive a flu shot. This includes:
 - Direct statements: "I don't get a flu shot", "I'm not getting the flu vaccine"
 - Habitual negatives: "I've never gotten a flu shot in my life", "I never get flu shots"
 - Dismissals or refusals: "no thanks", "I'll pass", "so I'm good" (in context of declining)
 - Past vaccination followed by explicit cessation: "I got the flu shot years ago but stopped getting them", "Got one once, never again"
 This category takes priority over "Previously-Vaccinated" when the author describes past vaccination but also makes clear they no longer get vaccinated.
- **Previously-Vaccinated:** The author mentions receiving a flu shot in a clearly prior flu season (e.g., "I got a flu shot last year", "back in 2019 I got vaccinated") with no indication of their current season status **AND** no indication they stopped getting vaccinated. If the author also expresses that they will not get one this year or have stopped, categorize as "Currently-Unvaccinated".
- **Possibly-Vaccinated:** The author expresses intent, consideration, plans, or obligation to get a flu shot but does not confirm they have actually received one. This includes:
 - Intent/plans: "thinking about getting a flu shot", "I need to get my flu shot", "getting my flu shot this week"
 - Habitual pattern (no specific event): "I get the flu shot every year", "I always get a flu shot", "I take the flu vaccine annually"
 - Obligation/mandate: "We have to get a flu shot for work", "I'm required to get the flu vaccine", "why do I have to take a flu vaccine"
 The distinction from Currently-Vaccinated is critical: habitual present-tense statements ("I get my flu shot every year") describe a **PATTERN**, not a specific completed event. Obligation statements describe a requirement, not a completed event. Both should be classified as Possibly-Vaccinated unless the tweet also describes a specific current-season vaccination event.
- **Other:** The tweet mentions flu vaccination but does NOT describe the author's own personal vaccination status. This includes: discussing someone else's vaccination, general advocacy or encouragement to vaccinate, sharing news or statistics, rhetorical or hypothetical statements, or any tweet where the author's own status cannot be determined.

Decision rules:

1. The classification must be based solely on what the author explicitly states about themselves – do not infer unstated information.
2. "Currently" refers to the present or most recent flu season. If the timing is ambiguous but the language is present-tense or recent, treat it as current.
3. If the tweet is **PRIMARILY** about a third party (child, family member, public figures) or is general commentary, classify as Other – **UNLESS** the author also reveals their own vaccination status in the tweet, in which case classify based on the author's status.

4. Habitual negatives ("I've never gotten one", "I never get flu shots") imply ongoing current status and should be classified as Currently-Unvaccinated.
5. If the tweet contains both past vaccination and current uncertainty, use the most specific information available.
6. **CRITICAL** – Habitual present tense vs. completed event:
 - "I get my flu shot every year" -> Possibly-Vaccinated (habitual pattern, no specific event confirmed)
 - "I got my flu shot" / "I got my flu shot today" -> Currently-Vaccinated (specific completed event)
 - "I'm getting my flu shot tomorrow" -> Possibly-Vaccinated (planned, not yet completed)
 - "Wore a mask to get a flu shot" -> Currently-Vaccinated (implies a specific completed visit)
7. Temporal references like "a few months ago", "in October", "last month" **WITHOUT** a prior-year reference should be treated as current season, not previously vaccinated.
8. Past vaccination + explicit cessation = Currently-Unvaccinated, not Previously-Vaccinated:
 - "I got the flu shot but stopped after that" -> Currently-Unvaccinated
 - "Got a flu vaccine once... never got another" -> Currently-Unvaccinated
9. Implicit refusals or dismissals count as Currently-Unvaccinated:
 - "so I'm good" (in context of not wanting a flu shot) -> Currently-Unvaccinated
 - "no thanks" / "I'll pass on the flu shot" -> Currently-Unvaccinated
10. Obligation/mandate language about the author implies Possibly-Vaccinated (intent through requirement), not Other:
 - "I have to get a flu shot for work" -> Possibly-Vaccinated
 - "We're required to get the flu vaccine" -> Possibly-Vaccinated

First reason through the tweet carefully, then provide your label.

// USER PROMPT

Tweet Created At: {timestamp}

Tweet: {raw_tweet_text}

JSON Schema (Subtask 1: Flu Vaccination Status Classification):

```
{
  "type": "json_schema",
  "name": "flu_vaccination_classification",
  "strict": true,
  "schema": {
    "type": "object",
    "properties": {
      "reasoning": {
        "type": "string",
        "description": "Step-by-step reasoning explaining which evidence in the tweet supports the chosen label, and why other labels were ruled out."
      },
      "label": {
        "type": "string",
        "enum": [
          "Currently-Vaccinated",
          "Currently-Unvaccinated",
          "Previously-Vaccinated",
          "Possibly-Vaccinated",
          "Other"
        ]
      },
      "description": "The flu vaccination status classification label."
    }
  },
  "required": [
    "reasoning",
    "label"
  ],
  "additionalProperties": false
}
```

```
}
```

Prompt Structure (Subtask 2: Flu Test Result):

```
// SYSTEM PROMPT
```

You are an expert annotator for a health informatics shared task. Your job is to classify social media posts (tweets) according to the author's self-reported flu test result or flu diagnosis.

Classify each tweet into exactly one of the following five categories:

- **Currently-Positive:** The author explicitly reports a recent positive flu test result or a current flu diagnosis (e.g., "I tested positive for flu", "just diagnosed with the flu", "I have the flu"). The result pertains to the current or very recent illness episode.
- **Currently-Negative:** The author explicitly reports a recent negative flu test result or a current diagnosis that excludes flu (e.g., "my flu test came back negative", "tested negative for flu", "they ruled out the flu"). The result pertains to the current or very recent illness episode.
- **Previously-Positive:** The author mentions having tested positive for flu or been diagnosed with flu during a clearly past season or episode, with no indication of a current test (e.g., "I had flu last January", "I was positive for flu back in 2019").
- **Previously-Negative:** The author mentions having tested negative for flu or been told they did not have flu during a clearly past episode, with no indication of a current test (e.g., "they tested me last year and flu came back negative").
- **Other:** The tweet mentions flu testing or diagnosis but does NOT describe the author's own personal test result. This includes: discussing someone else's test, general commentary about flu testing practices, encouraging others to get tested, reporting statistics, ambiguous or unresolvable statements, or any tweet where the author's own test result cannot be determined.

Decision rules:

1. The classification must be based solely on what the author explicitly states about their own flu test or diagnosis – do not infer unstated results.
2. "Currently" refers to a recent, ongoing, or just-concluded illness episode. Past-tense language describing events in a prior season should be classified as Previously-Positive or Previously-Negative.
3. If the tweet discusses someone else's test result (child, family, public figures) or is general commentary, classify as Other.
4. Symptoms alone (fever, cough) without a confirmed test result or diagnosis do not constitute a test result – classify as Other.
5. If multiple flu tests are mentioned (current and past), classify based on the most recent one.
6. A positive COVID test or other non-flu diagnosis does not qualify as a flu test result – classify as Other unless flu is also explicitly mentioned.

First reason through the tweet carefully, then provide your label.

```
// USER PROMPT
```

```
Tweet Created At: {timestamp}  
Tweet: {raw_tweet_text}
```

JSON Schema (Subtask 2: Flu Test Result Classification):

```
{  
  "type": "json_schema",  
  "name": "flu_test_result_classification",  
  "strict": true,  
  "schema": {  
    "type": "object",  
    "properties": {  
      "reasoning": {  
        "type": "string",  
        "description": "Step-by-step reasoning explaining which evidence in the tweet supports the chosen label, and why other labels were ruled out."  
      },  
      "label": {  
        "type": "string",  
        "enum": [  
          "Currently-Positive",  
          "Currently-Negative",  
          "Previously-Positive",  
          "Previously-Negative",  
          "Other"  
        ]  
      },  
      "description": "The flu test result classification label."  
    }  
  },  
  "required": [  
    "reasoning",  
    "label"  
  ],  
  "additionalProperties": false  
}
```

JSON Schema (Subtask 2: Flu Test Result Classification):

```
{  
  "type": "json_schema",  
  "name": "flu_test_result_classification",  
  "strict": true,  
  "schema": {  
    "type": "object",  
    "properties": {  
      "reasoning": {  
        "type": "string",  
        "description": "Step-by-step reasoning explaining which evidence in the tweet supports the chosen label, and why other labels were ruled out."  
      },  
      "label": {  
        "type": "string",  
        "enum": [  
          "Currently-Positive",  
          "Currently-Negative",  
          "Previously-Positive",  
          "Previously-Negative",  
          "Other"  
        ]  
      },  
      "description": "The flu test result classification label."  
    }  
  },  
  "required": [  
    "reasoning",  
    "label"  
  ],  
  "additionalProperties": false  
}
```

A.6 Task 4: Dial2Note Prompts and Schema

Prompt Structure:

```
// SYSTEM PROMPT
```

You are a clinical documentation specialist. Given a doctor-patient dialogue, generate a structured SOAP note (Subjective, Objective, Assessment, Plan).

GROUNDING RULE – every statement in the note must be directly supported by the dialogue:

- Demographics (name, age, sex, ethnicity): include ONLY if explicitly stated. Do not infer from names or voice.
- Doctor/specialist names: NEVER fabricate. Write "Refer to [specialty]" unless the doctor names them.
- Diagnoses: use only diagnoses the doctor explicitly states.

Instructions:

- Subjective: Document CC, HPI, ROS, PMH, surgical/family/social history, allergies, and medications. Keep the HPI concise – summarize in 2–4 sentences, don't restate every exchange. Include severity in the CC (e.g. "Moderate to severe shortness of breath").
- Objective: Record vital signs, physical exam, and test results. Use the doctor's exact clinical language for exam findings. When a vital is "normal" without a number, write "Normal (specific value not provided)".
- Assessment: Use a bulleted list of diagnoses. Include the doctor's clinical reasoning when stated. Include differential diagnoses if the doctor discusses them. Include ICD-10 codes if the doctor mentions them.
- Plan: Group items under bold sub-headings. Common headings: ****Medical Treatment:****, ****Medications:****, ****Imaging:****, ****Investigations:****, ****Referrals:****, ****Lifestyle Modifications:****, ****Follow-up:****, ****Patient Education and Counseling:****, ****Patient Agreements:****. Under each heading, list items as indented sub-bullets. Use the doctor's own words. Only include items explicitly discussed.
- When the patient explicitly agrees with the plan or the doctor confirms understanding, include ****Patient Agreements:**** as a final plan block.

Formatting rules:

- Use null for fields not mentioned; empty arrays [] for empty lists.
- For ROS and physical exam entries, use only canonical system names from the enum.
- Write `current_medications` as prose strings (e.g. "Metformin 500 mg twice daily for diabetes").
- Each plan item in the JSON array should be a category block: bold heading (e.g. ****Medications:****) followed by newline-indented sub-items ("\\n - Clindamycin 300 mg orally every 6 hours.\\n").
- Social history = patient's reported habits, NOT doctor's advice.
- ROS = only systems the doctor asked about or the patient reported symptoms for.
- Be concise but complete.

Below are three example dialogues and their expected SOAP notes. Note how each note includes ONLY information grounded in the dialogue.

--- EXAMPLE 1 (Rheumatology – Joint Pain) ---

Dialogue:

[doctor]: Hi, how are you doing today? What brings you in?

[patient]: Hi, doctor. I've been having some pretty bad joint pain, stiffness, and swelling lately.

[doctor]: I'm sorry to hear that. Can you tell me a bit more about these symptoms? How long have you been experiencing them?

[patient]: It's been about 6 months now. The pain is there every day and it's worse in the mornings. It's really affecting my ability to type and use a mouse at work.

[doctor]: I see. That sounds quite uncomfortable. Is there any redness or restricted range of motion in your joints?

[patient]: Yes, occasionally there's redness and my movement is definitely restricted. It's made my daily exercise routine almost impossible.

[doctor]: That sounds challenging. Have you noticed any other symptoms like fever or weight loss?

[patient]: No, no fever or weight loss.

[doctor]: Hmm, okay. How about your cardiovascular health? Any known issues there?

[patient]: I do have hypertension, but it's currently managed with medication.

[doctor]: Alright. And any skin issues?

[patient]: Yes, I have psoriasis. I'm managing it with topical corticosteroids.

[doctor]: Thank you for sharing that. Let's go over your vital signs. Your blood pressure is 130/85 mmHg, heart rate is 70 bpm, respiratory rate is 16 breaths per minute, and your temperature is 98.6 degrees F.

[patient]: Sounds normal, I guess?

[doctor]: Yes, those are within normal ranges. Now, let's move on to the physical examination. I'll check your joints and skin.

[patient]: Okay.

[doctor]: I see tenderness and swelling in your wrists, knees, and fingers. There's reduced range of motion in these joints, but no deformities or subluxations. Your skin has scaly plaques consistent with psoriasis.

[patient]: That matches what I've been feeling.

[doctor]: We'll need to do some investigations, including X-rays of the affected joints and blood work—Complete Blood Count (CBC), C-Reactive Protein (CRP), Erythrocyte Sedimentation Rate (ESR), and Rheumatoid Factor (RF).

[patient]: Alright.

[doctor]: Based on your symptoms and history, the primary diagnosis is Psoriatic Arthropathy. We also need to consider your psoriasis and hypertension.

[patient]: Okay, what's the plan?

[doctor]: For pain relief, I'll prescribe Naproxen 500 mg, to be taken orally twice daily. To slow the progression of your joint issues, we'll start Methotrexate 10 mg, taken orally once a week. You'll also need to take Folic acid 1 mg daily to counteract potential side effects of Methotrexate.

[patient]: Got it. Anything else?

[doctor]: I'll refer you to a physical therapist, Dr. Allison Smith, for joint mobility exercises and pain management techniques.

[patient]: That sounds helpful.

[doctor]: We'll need to schedule a follow-up appointment in 4 weeks to monitor how the medications are working and check for any side effects. Also, we'll order liver function tests every 6 weeks to ensure Methotrexate is safe for you.

[patient]: Okay, I understand.

[doctor]: Do you have any questions or concerns right now?

[patient]: Not at the moment. Just want to get this under control.

[doctor]: That's completely understandable. Make sure to follow up in 4 weeks and contact the clinic if you experience any adverse effects or worsening of symptoms.

[patient]: Will do. Thanks, doctor.

[doctor]: You're welcome. Take care and see you in 4 weeks.

[patient]: Thanks, you too.

Expected note:

```
{
  "subjective": {
    "chief_complaint": "Joint pain, stiffness, and swelling for 6 months",
    "hpi": "The patient presents with joint pain, stiffness, and swelling for 6 months. The pain is daily and worse in the mornings, impacting ability to type and use a mouse at work. Occasional redness and restricted range of motion noted, hindering daily exercise.",
    "ros": [
      {
        "system": "Musculoskeletal",
        "finding": "Positive for joint pain, swelling, stiffness, occasional redness, and restricted range of motion"
      },
      {
        "system": "Constitutional",
        "finding": "Denies fever, weight loss"
      }
    ]
  }
}
```

```

    {"system": "Cardiovascular", "finding": "Known
      history of hypertension, currently
      managed with medication"},
    {"system": "Skin", "finding": "Positive for
      psoriasis, currently managed with topical
      corticosteroids"}
  ],
  "past_medical_history": ["Hypertension", "
    Psoriasis"],
  "surgical_history": [],
  "family_history": [],
  "social_history": null,
  "allergies": [],
  "current_medications": ["Topical corticosteroids
    for psoriasis", "Antihypertensive
    medication (unspecified)"]
},
"objective": {
  "vital_signs": {
    "blood_pressure": "130/85 mmHg",
    "heart_rate": "70 bpm",
    "respiratory_rate": "16 breaths per minute",
    "temperature": "98.6 F",
    "oxygen_saturation": null,
    "weight": null, "height": null, "bmi": null, "
    pain_scale": null
  },
  "physical_exam": [
    {"system": "Musculoskeletal", "finding": "
      Tenderness and swelling observed in
      multiple joints including wrists, knees,
      and fingers. Reduced range of motion
      noted in the affected joints. No
      deformities or subluxations detected."},
    {"system": "Skin", "finding": "Scaly plaques
      consistent with psoriasis"}
  ],
  "test_results": []
},
"assessment": "- Psoriatic Arthropathy\n-
  Psoriasis\n- Hypertension",
"plan": [
  "**Medical Treatment:**\n - Prescribe Naproxen
    500 mg, oral, twice daily for pain relief.\n
    - Start Methotrexate 10 mg, oral, once
    weekly to slow disease progression.\n -
    Advise taking Folic acid 1 mg, oral, daily
    to mitigate Methotrexate side effects.",
  "**Investigations:**\n - Order X-ray of
    affected joints (wrists, knees, fingers).\n
    - Order blood work: Complete Blood Count
    (CBC), C-Reactive Protein (CRP),
    Erythrocyte Sedimentation Rate (ESR),
    Rheumatoid Factor (RF).",
  "**Referrals:**\n - Refer to physical therapist
    for joint mobility exercises and pain
    management techniques.",
  "**Follow-up:**\n - Schedule follow-up
    appointment in 4 weeks to monitor
    medication efficacy and any potential side
    effects.\n - Order liver function tests
    every 6 weeks to ensure the safety of
    Methotrexate.",
  "**Patient Agreements:**\n - The patient
    understands and agrees with the recommended
    medical treatment plan.\n - Follow up in
    4 weeks for evaluation.\n - Contact the
    clinic if experiencing any adverse effects
    or worsening of symptoms."
]
}

```

--- EXAMPLE 2 (Obstetrics – Labor with Meconium) ---

Dialogue:

[doctor]: Good morning, how are you feeling today?
 [patient]: Good morning, Doctor. I'm feeling a bit
 anxious, to be honest.
 [doctor]: I understand. Can you tell me a bit more
 about what brought you in today?
 [patient]: Well, I'm in labor and my water broke.
 The fluid was a greenish color, and I got
 really worried.
 [doctor]: I see. How long ago did your labor start?
 [patient]: About 12 hours ago. I had some clear and
 bloody show before my water broke.

[doctor]: Okay, thank you for sharing that. Have you
 experienced any fever or chills?
 [patient]: No, none at all.
 [doctor]: How about nausea or vomiting during labor?
 [patient]: No, I haven't had any nausea or vomiting.
 [doctor]: Any shortness of breath?
 [patient]: No, my breathing has been normal.
 [doctor]: And how about urination? Any burning or
 discomfort?
 [patient]: No, my urination has been normal.
 [doctor]: Any significant pain aside from the labor
 contractions?
 [patient]: No, just the usual labor pains.
 [doctor]: Alright, let's review your vital signs.
 Your blood pressure is 120/80 mmHg, heart rate
 is 85 bpm, temperature is 98.6 degrees F, and
 respiratory rate is 18 breaths/min. All stable.
 [patient]: That's good to hear.
 [doctor]: On physical examination, your fundal
 height is consistent with term pregnancy and
 the baby is in a vertex position. Your cervix
 is fully dilated and 100% effaced. We did
 notice decelerations on the continuous
 electronic fetal monitoring, but we managed
 them with maternal positional changes and
 supplemental oxygen.
 [patient]: Okay, that sounds a bit concerning.
 [doctor]: Yes, and the greenish discoloration of the
 amniotic fluid indicates meconium staining,
 which suggests fetal stress.
 [patient]: What does that mean for the baby?
 [doctor]: The presence of meconium in the amniotic
 fluid can potentially lead to aspiration and
 respiratory complications for the newborn. We
 will need to take immediate action once the
 baby is born.
 [patient]: What kind of actions?
 [doctor]: We will need to suction the infant's
 airway immediately at birth to clear any
 meconium that might be present. The newborn
 will also be admitted to the Neonatal Intensive
 Care Unit (NICU) for close monitoring of their
 respiratory status.
 [patient]: That sounds serious. Will the baby need
 oxygen?
 [doctor]: If necessary, we will provide supplemental
 oxygen to manage any respiratory distress. We
 will also refer to Neonatology for
 comprehensive evaluation and management.
 [patient]: Is there anything I need to do?
 [doctor]: After delivery, we will have a pediatric
 follow-up within one week post-discharge to
 monitor for any delayed respiratory symptoms or
 other complications. We will also continue to
 monitor and manage your gestational diabetes
 through the postpartum period.
 [patient]: Okay, what should I look out for in the
 baby?
 [doctor]: It's important to recognize signs of
 respiratory distress in the newborn, such as
 cyanosis, rapid breathing, or grunting. I will
 discuss this in more detail with you and your
 partner.
 [patient]: Alright, I'll keep an eye out.
 [doctor]: We will also discuss maintaining a
 balanced diet and monitoring blood glucose
 levels post-delivery. Mild exercise routines
 postpartum can aid in recovery, but we will
 provide specific advice based on your condition

[patient]: That sounds good.
 [doctor]: Do you have any questions or concerns?
 [patient]: No, I think you've covered everything.
 Thank you for explaining it all.
 [doctor]: You're welcome. Our priority is the health
 and safety of both you and your baby. We'll be
 here every step of the way. Let's get you
 ready for delivery.
 [patient]: Thank you, Doctor. I appreciate it.
 [doctor]: You're welcome. Let's proceed and ensure
 everything goes smoothly.

Expected note:

```

{
  "subjective": {
    "chief_complaint": "Labor and delivery
      complicated by meconium in amniotic fluid",
  }
}

```

```

"hip": "The patient is presenting with labor at term. She noticed rupture of membranes accompanied by greenish discoloration of the amniotic fluid, indicative of meconium staining. Labor onset was 12 hours ago with clear and bloody show prior to membrane rupture.",
"ros": [
  {"system": "Constitutional", "finding": "No fever or chills"},
  {"system": "Gastrointestinal", "finding": "No nausea or vomiting during labor"},
  {"system": "Respiratory", "finding": "No shortness of breath"},
  {"system": "Genitourinary", "finding": "Normal urination, no dysuria"},
  {"system": "Musculoskeletal", "finding": "No significant pain other than labor contractions"}
],
"past_medical_history": ["Gestational diabetes"],
"surgical_history": [],
"family_history": [],
"social_history": null,
"allergies": [],
"current_medications": []
},
"objective": {
  "vital_signs": {
    "blood_pressure": "120/80 mmHg",
    "heart_rate": "85 bpm",
    "respiratory_rate": "18 breaths/min",
    "temperature": "98.6 F",
    "oxygen_saturation": null,
    "weight": null, "height": null, "bmi": null, "pain_scale": null
  },
  "physical_exam": [
    {"system": "Abdomen", "finding": "Fundal height consistent with term pregnancy, fetal position vertex"},
    {"system": "Pelvic", "finding": "Cervix fully dilated, 100% effaced"}
  ],
  "test_results": ["Continuous electronic fetal monitoring: decelerations noted, managed with maternal positional changes and supplemental oxygen", "Amniotic fluid: positive for meconium staining"]
},
"assessment": "- Labor and delivery complicated by meconium in amniotic fluid.\n- Medical Reasoning: The presence of meconium in the amniotic fluid suggests fetal stress, which could lead to aspiration and potential respiratory complications for the newborn.",
"plan": [
  "**Medical Treatment:**\n - Immediate suctioning of the infant's airway at birth to clear any meconium present.\n - Admission of the newborn to the Neonatal Intensive Care Unit (NICU) for close monitoring of respiratory status.\n - Supplemental oxygen to the newborn if necessary to manage any respiratory distress.",
  "**Referrals:**\n - Referral to Neonatology for comprehensive evaluation and management of potential aspiration and respiratory complications.",
  "**Follow-up:**\n - Pediatric follow-up within 1 week post-discharge to monitor for delayed respiratory symptoms or other complications.\n - Continued monitoring and management of gestational diabetes through the postpartum period.",
  "**Patient Education and Counseling:**\n - Discussed with the patient the importance of recognizing signs of respiratory distress in the newborn, such as cyanosis, rapid breathing, or grunting.\n - Discussed maintaining a balanced diet and monitoring blood glucose levels post-delivery.\n - Encouraged mild exercise routines postpartum to aid in recovery.",

```

```

**Patient Agreements:**\n - The patient understands and agrees with the recommended medical treatment plan."
]
}

```

--- EXAMPLE 3 (Emergency – Sports Injury) ---

Dialogue:

```

[doctor]: Hi there, I understand you're experiencing some pain and swelling in your right lower leg. Can you tell me what happened?
[patient]: Yes, I was playing in a local sports league game about two hours ago and I hurt my leg. It's been hurting and swelling up since then.
[doctor]: Oh, that sounds painful. Can you describe the pain for me? How severe is it on a scale of 1 to 10?
[patient]: It's pretty bad, I'd say it's around a 6 or 7. It hurts a lot, and I can't walk or put any weight on it.
[doctor]: I see. Did you notice any bruising or other symptoms?
[patient]: Yes, there's some slight bruising and it's swollen. The pain has been continuous since the injury.
[doctor]: Alright, let's go over some other symptoms. Have you had any fever or chills?
[patient]: No, none of that.
[doctor]: How about chest pain or palpitations?
[patient]: No, my chest feels fine.
[doctor]: Any shortness of breath or cough?
[patient]: No, nothing like that.
[doctor]: Aside from the pain and swelling in your right lower leg, are you experiencing any other joint pain or muscle weakness?
[patient]: No, just the leg.
[doctor]: Okay, thank you for that information. Let's take a look at your vital signs. Your blood pressure is 122 over 78, heart rate is 78 beats per minute, respiratory rate is 16 breaths per minute, temperature is 98.6 degrees Fahrenheit, and your oxygen saturation is 98% on room air.
[patient]: Okay, that sounds good, I guess.
[doctor]: Now, let's do a physical examination. I'll start with listening to your heart and lungs. Your cardiovascular system shows a regular rate and rhythm with no murmurs. Your lungs are clear to auscultation bilaterally.
[patient]: Alright.
[doctor]: I'm going to examine your right lower leg now. I see there's swelling and tenderness. How does it feel when I press here?
[patient]: Ouch, that hurts.
[doctor]: Your range of motion is limited due to pain. I don't see any obvious deformities, and there's no crepitus or instability noted. We should get an X-ray to rule out any fractures or other significant injuries. Does that sound okay?
[patient]: Yes, that's fine.
[doctor]: While we wait for the X-ray results, let's discuss the plan. I recommend applying ice to your leg every 2 hours for the next 48 hours to reduce swelling. Elevate your leg above heart level as much as possible.
[patient]: Alright, I can do that.
[doctor]: Also, avoid putting weight on your leg. I'll provide you with crutches. For the pain, I'll prescribe ibuprofen 400 mg tablets. Take one every 6 hours as needed, up to 30 tablets.
[patient]: Got it, thanks.
[doctor]: Make sure to follow up with your primary care physician in one week for reassessment. If you experience increased pain, swelling, redness, or if you cannot move your leg, return to the emergency department immediately.
[patient]: Okay, I'll keep an eye on it.
[doctor]: Do you have any questions about the instructions?
[patient]: No, I think I got it. Thank you.
[doctor]: Great. We'll wait for the X-ray results and then proceed accordingly. Take care and rest that leg.
[patient]: Thank you, Doctor.

```

[doctor]: You're welcome. I'll be back shortly once we have the X-ray results.

Expected note:

```
{
  "subjective": {
    "chief_complaint": "Moderate pain and swelling in the right lower leg",
    "hpi": "The patient presents with pain and swelling in the right lower leg. The injury occurred while participating in a local sports league game about 2 hours ago. Reports continuous pain since the injury, accompanied by swelling and slight bruising. Pain severity is 6-7/10. Unable to walk or bear weight on the affected leg.",
    "ros": [
      {"system": "Constitutional", "finding": "Denies fever, chills"},
      {"system": "Cardiovascular", "finding": "Denies chest pain, palpitations"},
      {"system": "Respiratory", "finding": "Denies shortness of breath, cough"},
      {"system": "Musculoskeletal", "finding": "Reports pain, swelling, and slight bruising in the right lower leg; denies other joint pain or muscle weakness"}
    ],
    "past_medical_history": [],
    "surgical_history": [],
    "family_history": [],
    "social_history": null,
    "allergies": [],
    "current_medications": []
  },
  "objective": {
    "vital_signs": {
      "blood_pressure": "122/78 mmHg",
      "heart_rate": "78 bpm",
      "respiratory_rate": "16 breaths per minute",
      "temperature": "98.6 F",
      "oxygen_saturation": "98% on room air",
      "weight": null, "height": null, "bmi": null, "pain_scale": "6-7/10"
    },
    "physical_exam": [
      {"system": "Cardiovascular", "finding": "Regular rate and rhythm, no murmurs"},
      {"system": "Respiratory", "finding": "Lungs clear to auscultation bilaterally"},
      {"system": "Extremities", "finding": "Right lower leg: swelling and tenderness noted, limited range of motion due to pain, no obvious deformities, no crepitus or instability noted"}
    ],
    "test_results": ["X-ray of right lower leg ordered to rule out fractures or other significant injuries; results pending"]
  },
  "assessment": "- Right lower leg injury with pain, swelling, and inability to bear weight.\n- Differential: Possible fracture vs soft tissue injury; X-ray ordered.",
  "plan": [
    "**Imaging:**\n - X-ray of right lower leg to rule out fractures or other significant injuries. Await and review results.",
    "**Medical Treatment:**\n - Ibuprofen 400 mg tablets, oral, every 6 hours as needed for pain; 30 tablets prescribed.\n - Apply ice to leg every 2 hours for the next 48 hours to reduce swelling.\n - Elevate leg above heart level as much as possible.\n - Avoid weight-bearing on the affected leg; crutches provided.",
    "**Follow-up Recommendations:**\n - Follow up with primary care physician in 1 week for reassessment.",
    "**Patient Education and Counseling:**\n - Patient advised to rest and restrict activities that could exacerbate the injury.\n - Return to the emergency department immediately for increased pain, swelling, redness, or inability to move the leg."
  ]
}
```

// USER PROMPT

Dialogue: {raw_dialogue_text}

JSON Schema:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "SOAPNoteProse",
  "description": "SOAP note with prose-style fields optimised for lexical overlap with MedSynth reference notes",
  "type": "object",
  "additionalProperties": false,
  "required": [
    "subjective",
    "objective",
    "assessment",
    "plan"
  ],
  "properties": {
    "subjective": {
      "$ref": "#/defs/Subjective"
    },
    "objective": {
      "$ref": "#/defs/Objective"
    },
    "assessment": {
      "type": "string",
      "description": "Clinical assessment. May be a single diagnosis line, a bulleted list of diagnoses, or a short narrative paragraph synthesising the findings - match whichever style the clinical context warrants. Include ICD-10 codes only if explicitly mentioned in the dialogue."
    },
    "plan": {
      "type": "array",
      "description": "Ordered list of plan items written in clinical prose. Each item should be a complete sentence or short paragraph. Prefix with a category label when helpful (e.g. 'Medication:', 'Imaging:', 'Referral:', 'Follow-up:'). Mirror the doctor's exact wording from the dialogue wherever possible.",
      "items": {
        "type": "string"
      }
    }
  },
  "defs": {
    "Subjective": {
      "type": "object",
      "additionalProperties": false,
      "required": [
        "chief_complaint",
        "hpi",
        "ros",
        "past_medical_history",
        "surgical_history",
        "family_history",
        "social_history",
        "allergies",
        "current_medications"
      ],
      "properties": {
        "chief_complaint": {
          "type": [
            "string",
            "null"
          ],
          "description": "Brief statement of the patient's presenting problem"
        },
        "hpi": {
          "type": [
            "string",
            "null"
          ],
          "description": "History of Present Illness - narrative paragraph. Always include the patient's full name, age, sex, and ethnicity when available in the dialogue."
        }
      }
    }
  }
}
```

```

    },
    "ros": {
      "type": "array",
      "description": "Review of Systems entries",
      "items": {
        "$ref": "#/$defs/ROSEntry"
      }
    },
    "past_medical_history": {
      "type": "array",
      "description": "List of pertinent current or past medical conditions",
      "items": {
        "type": "string"
      }
    },
    "surgical_history": {
      "type": "array",
      "description": "Prior surgeries as prose strings, e.g. 'Appendectomy (2018)'",
      "items": {
        "type": "string"
      }
    },
    "family_history": {
      "type": "array",
      "description": "Family history as prose strings, e.g. 'Father: hypertension, diabetes'",
      "items": {
        "type": "string"
      }
    },
    "social_history": {
      "type": [
        "string",
        "null"
      ],
      "description": "Social history as a short prose paragraph or semicolon-separated summary covering tobacco, alcohol, drug use, diet, exercise, occupation, etc. Use null if not discussed."
    },
    "allergies": {
      "type": "array",
      "description": "Free-text allergy strings, e.g. 'Penicillin - rash'",
      "items": {
        "type": "string"
      }
    },
    "current_medications": {
      "type": "array",
      "description": "Current medications as prose strings mirroring the dialogue wording, e.g. 'Metformin 500 mg twice daily for diabetes'",
      "items": {
        "type": "string"
      }
    }
  },
  "Objective": {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "vital_signs",
      "physical_exam",
      "test_results"
    ],
    "properties": {
      "vital_signs": {
        "anyOf": [
          {
            "$ref": "#/$defs/VitalSigns"
          },
          {
            "type": "null"
          }
        ]
      },
      "physical_exam": {
        "type": "array",
        "description": "Physical exam findings by body system or region",
        "items": {
          "$ref": "#/$defs/PhysicalExamEntry"
        }
      },
      "test_results": {
        "type": "array",
        "description": "Completed test results as prose strings, e.g. 'Chest X-ray: No acute findings'",
        "items": {
          "type": "string"
        }
      }
    }
  },
  "ROSEntry": {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "system",
      "finding"
    ],
    "properties": {
      "system": {
        "type": "string",
        "enum": [
          "Constitutional",
          "Cardiovascular",
          "Respiratory",
          "Gastrointestinal",
          "Genitourinary",
          "Musculoskeletal",
          "Neurological",
          "Psychiatric",
          "Endocrine",
          "Skin",
          "HEENT",
          "Eyes",
          "ENT",
          "Hematologic",
          "Allergic",
          "Breast",
          "Gynecological",
          "Sleep",
          "Other"
        ]
      },
      "finding": {
        "type": "string",
        "description": "Clinical finding or symptom report for this system"
      }
    }
  },
  "PhysicalExamEntry": {
    "type": "object",
    "additionalProperties": false,
    "required": [
      "system",
      "finding"
    ],
    "properties": {
      "system": {
        "type": "string",
        "enum": [
          "General",
          "HEENT",
          "Neck",
          "Cardiovascular",
          "Respiratory",
          "Abdomen",
          "Musculoskeletal",
          "Neurological",
          "Skin",
          "Extremities",
          "Psychiatric",
          "Genitourinary",
          "Pelvic",
          "Rectal",
          "Breast",
          "Eyes"
        ]
      },
      "finding": {
        "type": "string",

```



```

        "description": "Objective examination
                        finding for this region"
    }
},
},
"VitalSigns": {
    "description": "All values stored as strings
                    to preserve units and qualifiers",
    "type": "object",
    "additionalProperties": false,
    "required": [
        "blood_pressure",
        "heart_rate",
        "respiratory_rate",
        "temperature",
        "oxygen_saturation",
        "weight",
        "height",
        "bmi",
        "pain_scale"
    ],
    "properties": {
        "blood_pressure": {
            "type": [
                "string",
                "null"
            ],
            "description": "e.g. '120/80 mmHg'"
        },
        "heart_rate": {
            "type": [
                "string",
                "null"
            ],
            "description": "e.g. '72 bpm'"
        },
        "respiratory_rate": {
            "type": [
                "string",
                "null"
            ],
            "description": "e.g. '16 breaths/min'"
        },
        "temperature": {
            "type": [
                "string",
                "null"
            ],
            "description": "e.g. '98.6 degrees F'"
        },
        "oxygen_saturation": {
            "type": [
                "string",
                "null"
            ],
            "description": "e.g. '98% on room air'"
        },
        "weight": {
            "type": [
                "string",
                "null"
            ]
        },
        "height": {
            "type": [
                "string",
                "null"
            ]
        },
        "bmi": {
            "type": [
                "string",
                "null"
            ]
        },
        "pain_scale": {
            "type": [
                "string",
                "null"
            ],
            "description": "e.g. '4/10'"
        }
    }
}
}
}
}

```

A.7 Task 5: Patient Metadata Prompts and Schema

Prompt Structure:

// SYSTEM PROMPT

You are an expert biomedical NLP system and epidemiological data auditor. Your task is to analyze sentences extracted from PubMed articles that report generating SARS-CoV-2 sequences, and determine if the sentence contains a valid mention of Patient Metadata for the study's own patients or samples.

DEFINITION OF PATIENT METADATA CATEGORIES:

"age", "sex", "race/ethnicity", "symptoms", "disease_severity", "viral_load", "duration_of_infection", "lab_results", "vital_signs", "treatments", "hospitalization", "outcomes", "comorbidities", "risk_factors", "vaccination_status", "place_of_residence", "geographic_location", "travel_history".

WHAT COUNTS AS "geographic_location" (this category is commonly misclassified):

POSITIVE – any of these patterns:

- The sentence names WHERE the study's own samples, sequences, or patients are from: "Indian SARS-CoV-2 sequences", "44 Hong Kong strains", "Maradi genomes", "Phylogenetic Analysis of SARS-CoV-2 in Colombia" (section title naming the study country)
- The sentence describes where samples were collected or received: "Nasopharyngeal swabs were received from the Civil Hospital Hisar, Haryana (India)"
- The sentence attributes sequences to a specific location in the current study: "residence-hall sequences clustered with viruses circulating in Dane County"
- The sentence reports testing or positivity rates for the study's geographic area: "10.8% overall positivity rate for the Central Texas geography"

NEGATIVE – any of these patterns:

- Regional case counts or epidemiological statistics not tied to specific patients: "307 infections in New South Wales"
- Where a variant/lineage was first globally detected: "B.1.1.7 was first detected in the UK"
- Phylogenetic comparison to EXTERNAL reference sequences from other countries: "more closely related to strains seen in South Korea (n = 7) and France (n = 5)" when South Korea and France are NOT the study's own sample sources
- Importation analysis at population level: "87% of imported cases came from Europe"

WHAT COUNTS AS "hospitalization":

POSITIVE: "cases admitted to our hospital", "patient was hospitalized", "required ICU admission"

NEGATIVE: "we enrolled patients from 4 hospitals" (study design, no hospitalization metadata)

CRITICAL DISTINCTION – PATIENT-LEVEL vs. POPULATION-LEVEL vs. STUDY DESIGN:

- Patient metadata must describe individuals or a defined cohort in the current study. NOT patient metadata:
- Population-level case statistics: "5,093 cases reported in Amazonas"
 - Epidemiological context: "the first confirmed case was reported in..."
 - Study enrollment descriptions without actual patient values: "we enrolled patients from 4 hospitals in Hong Kong"
 - Study protocols: "participants who tested positive were placed in isolation"

CLASSIFICATION TRAPS – if a sentence matches a trap, it MUST be labeled NEGATIVE:

TRAP A (Previous/External Studies): Metadata from other studies, prior publications, external databases, or reference sequences.

Examples: "a postmortem pathological study found ...", "consistent with previously published evidence (Author, Year)"

TRAP B (Methodology/Variables): Variables or parameters mentioned without actual values for patients.

Examples: "we analyzed how severity was impacted by age" (no ages given), "the lack of association between G614 and hospitalization" (statistical finding about a variable, not actual patient data)

EXCEPTION: Figure/table captions that describe the study's own cohort data (e.g., "persistence of symptoms at 2 and 6 months after hospital admission for COVID-19", "Demographic and clinical characteristics of patients") -> POSITIVE.

EXCEPTION: Sentences describing sample collection that imply hospitalization (e.g., "collected from 126 cases admitted to our hospital") -> POSITIVE.

TRAP C (General Virology): Viral genetics, mutations, phylogenetics, or lab methods WITHOUT patient-level details.

CRITICAL: TRAP C should ONLY be triggered when the sentence is PURELY about the virus with NO reference to the study's own patients or sample origins. If the sentence mentions ANY of the following tied to the current study, do NOT trigger TRAP C - label POSITIVE instead:

- Geographic origin of the study's samples or sequences (country, city, region, hospital)
- A treatment given to the study's patients (e.g., "convalescent plasma therapy")
- Patient sample types or collection context that implies clinical metadata

If uncertain whether geography refers to the study's own samples vs. external references, consider it the study's own and label POSITIVE.

LOGICAL CONSTRAINT (MANDATORY):

If is_trap_triggered is true or trap_type is not "None", then patient_metadata_mentioned MUST be false. These fields must be consistent.

LABEL POSITIVE (TRUE):

- The sentence reports or references patient metadata for the current study. This includes:
- Actual demographic values (ages, sex distribution)
 - Clinical information (symptoms, severity, outcomes, treatments, hospitalization)
 - Geographic origin of the study's samples or patients
 - Lab results for the study's patients (test results, Ct values, viral load)
 - Sample collection context that implies location or clinical setting
 - Section/figure/table titles that name the study's geography or cohort characteristics

LABEL NEGATIVE (FALSE):

The sentence has no valid patient metadata for the study's own patients, or it falls into a trap above.

// USER PROMPT

Analyze the following PubMed extract. Apply your classification rules to determine if any patient metadata is mentioned, and extract the required information.

Extract: {raw_pubmed_text}

JSON Schema:

```
{
  "name": "covid_19_metadata_labeling",
  "strict": true,
  "schema": {
    "type": "object",
    "required": [
      "unique_id",
      "patient_metadata"
    ],
    "additionalProperties": false,
    "properties": {
```

```
    "unique_id": {
      "type": "string",
      "description": "The unique identifier as provided in the text"
    },
    "patient_metadata": {
      "type": "object",
      "required": [
        "patient_metadata_mentioned",
        "patient_metadata_citation"
      ],
      "additionalProperties": false,
      "properties": {
        "patient_metadata_mentioned": {
          "type": "boolean",
          "description": "Whether any patient metadata clearly associated with the study's sequenced cases is mentioned in the text"
        },
        "patient_metadata_citation": {
          "type": "string",
          "description": "The verbatim input text that supports the label for patient_metadata_mentioned. If patient_metadata_mentioned is false, this should be an empty string."
        }
      }
    }
  }
}
```

A.8 Task 6: TNM Extraction Prompts and Schema

Prompt Structure:

// SYSTEM PROMPT

You are an expert oncological pathologist. Your task is to extract the TNM cancer stage from TCGA pathology reports (text provided)

You must analyze the provided clinical text and independently determine three specific classifications:

1. T (Primary Tumor Stage): Represents the size and extent of the main tumor.
 2. N (Lymph Node Involvement): Represents the number and location of regional lymph nodes that have cancer.
 3. M (Metastasis): Represents whether the cancer has spread to other parts of the body.
- If explicit staging is missing, use the macroscopic and microscopic descriptions (e.g., tumor size in cm, number of positive lymph nodes) to infer the correct clinical stage based on standard oncological guidelines.

<Critical: Metastasis Detection>

DO NOT reflexively default to M0. Carefully search the ENTIRE report for ANY of the following M1 indicators:

- Explicit "M1" or "pM1" notation anywhere in the text
- Tumor found in a distant organ (e.g., liver metastasis, bone metastasis, brain metastasis, contralateral lung)
- Phrases like "distant metastasis", "metastatic deposit in [organ]", "consistent with metastatic [cancer type]"
- Tumor in non-regional lymph node stations
- Peritoneal carcinomatosis, pleural metastasis, omental deposits
- Reports describing a biopsy of a SECONDARY site (not the primary tumor)

Only assign M0 if you have thoroughly checked and found NO evidence of distant spread.

</Critical>

// USER PROMPT

READ CAREFULLY: Pathology reports are often surgical and may explicitly stage the tumor (e.g., "pT2 N1 Mx").

- If explicit staging is missing, use the macroscopic and microscopic descriptions (e.g., tumor size in cm, number of positive lymph nodes) to infer the correct clinical stage based on standard oncological guidelines.

JSON Schema:

```
{
  "name": "TNM_staging_reason_first",
  "strict": true,
  "schema": {
    "type": "object",
    "required": [
      "patient_filename",
      "primary_tumour_stage",
      "lymph_node_involvement",
      "metastasis"
    ],
    "additionalProperties": false,
    "properties": {
      "patient_filename": {
        "type": "string",
        "description": "The patient filename as provided in the text"
      },
      "primary_tumour_stage": {
        "type": "object",
        "required": [
          "evidence",
          "explicit_tnm_string",
          "reasoning",
          "T_stage"
        ],
        "additionalProperties": false,
        "properties": {
          "evidence": {
            "type": "array",
            "items": {
              "type": "string",
              "description": "Verbatim quotes from the report relevant to T staging (tumor size, depth of invasion, organ involvement, explicit pT notation)."
            },
            "description": "Extract ALL verbatim text from the report relevant to determining T stage BEFORE making a classification."
          },
          "explicit_tnm_string": {
            "type": "string",
            "description": "If the report contains an explicit T stage (e.g. 'pT2', 'T3', 'Stage IIIA'), copy it here verbatim. If none found, write 'none'."
          },
          "reasoning": {
            "type": "string",
            "description": "Based on the evidence and any explicit staging, explain your reasoning for the T stage classification."
          },
          "T_stage": {
            "type": "string",
            "enum": [
              "T1",
              "T2",
              "T3",
              "T4"
            ],
            "description": "Your final T stage classification."
          }
        },
        "description": "Your final T stage classification."
      },
      "lymph_node_involvement": {
        "type": "object",
        "required": [
          "evidence",
          "explicit_tnm_string",
          "lymph_node_counts",
          "reasoning",
          "N_stage"
        ],
        "additionalProperties": false,
        "properties": {
          "evidence": {
            "type": "array",
            "items": {
              "type": "string",
              "description": "Verbatim quotes from the report about lymph nodes (counts, stations, positivity, explicit pN notation)."
            },
            "description": "Extract ALL verbatim text from the report relevant to determining N stage BEFORE making a classification."
          },
          "explicit_tnm_string": {
            "type": "string",
            "description": "If the report contains an explicit N stage (e.g. 'pN1', 'N2'), copy it here verbatim. If none found, write 'none'."
          },
          "lymph_node_counts": {
            "type": "string",
            "description": "Summarize the lymph node counts: how many positive out of how many examined? e.g. '3/15 positive'. Write 'not reported' if no counts given."
          },
          "reasoning": {
            "type": "string",
            "description": "Based on the evidence, explicit staging, and node counts, explain your reasoning for the N stage classification."
          },
          "N_stage": {
            "type": "string",
            "enum": [
              "N0",
              "N1",
              "N2",
              "N3"
            ],
            "description": "Your final N stage classification."
          }
        },
        "description": "Your final N stage classification."
      },
      "metastasis": {
        "type": "object",
        "required": [
          "evidence_for_m1",
          "evidence_against_m1",
          "explicit_tnm_string",
          "reasoning",
          "M_stage"
        ],
        "additionalProperties": false,
        "properties": {
          "evidence_for_m1": {
            "type": "array",
            "items": {
              "type": "string",
              "description": "Verbatim quotes suggesting distant metastasis (distant organ involvement, pM1, non-regional nodes, secondary site biopsies)."
            },
            "description": "Extract ALL verbatim text that could indicate M1. If nothing found, return empty array."
          },
          "evidence_against_m1": {
            "type": "array",
            "items": {
              "type": "string",
              "description": "Verbatim quotes suggesting no distant metastasis (negative distant workup, pM0, confined to primary site)."
            },
            "description": "Extract ALL verbatim text that could indicate M1. If nothing found, return empty array."
          },
          "explicit_tnm_string": {
            "type": "string",
            "description": "If the report contains an explicit M stage (e.g. 'pM1', 'M2'), copy it here verbatim. If none found, write 'none'."
          },
          "reasoning": {
            "type": "string",
            "description": "Based on the evidence, explicit staging, and node counts, explain your reasoning for the M stage classification."
          },
          "M_stage": {
            "type": "string",
            "enum": [
              "M0",
              "M1",
              "M2",
              "M3"
            ],
            "description": "Your final M stage classification."
          }
        },
        "description": "Your final M stage classification."
      }
    }
  }
}
```

